

Date	29 September 2026
Team ID	6ab2a427dbd693aad5a7d4b
Project Name	Student Enrollment & Course Management System (SECMS)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Secure login with Student ID/Email & password, OTP, Google SSO, password reset, MFA for Admin. Session timeout 15 min.	High
2	Authorization levels	Student: view/apply/pay, Faculty: manage courses & students, Admin: full access. Role-based API restriction.	High
3	External interfaces	Payment Gateway Razorpay/UPI, Email/SMS SendGrid, University DB API, AWS S3 for docs storage.	High
4	Transactions processing	Payment & enrollment ACID transaction. Fail means rollback. Real-time seat locking to avoid overbooking. Audit log.	High
5	Reporting	Admin: enrolled per course, fee, dropout, faculty workload. Export	Medium

		PDF/Excel. Student: status & receipt.	
6	Business rules, etc.	Max 6 courses/sem, Prerequisite check, FCFS seat, 24hr payment rule, 75% attendance rule.	High
7	Compliance to laws or regulations	GDPR & DPDP Act 2023, PCI-DSS for payments, AES-256 encryption, 5 years retention per UGC.	High
8	Other	Support 1000+ users, <2 sec response, 99.9% uptime, Daily backup, Mobile responsive, Audit log, Helpdesk chatbot.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	System must load pages quickly, process enrollment and payment without delay even during peak admission.	Page loads <2 sec, API <500ms, Enrollment <3 sec, 500 searches/sec
2	Scalability	Handle growth in users/courses during admission season. Support vertical & horizontal scaling.	10,000 concurrent users, 50,000+ student records, Auto-scale when load >70%
3	Security & Data Privacy	Protect student data, passwords, payment info. Prevent SQL injection, XSS, CSRF. Role-based audit logs.	AES-256, SSL/TLS, bcrypt password, PCI-DSS, OTP, Audit log
4	Reliability & Availability	Available 24/7 during admission, fault tolerance, quick recovery, no data loss. Daily backup.	99.9% uptime/month, Failover, Daily backup 2 AM, RTO <15 min, RPO <1 hr
5	Usability & Accessibility	Easy 3-step enrollment, mobile responsive, accessible for disabled, multi-language.	WCAG 2.1 AA, Mobile responsive, <3 clicks enrollment, English/Tamil, Chatbot

6	Other	Maintainability, Portability, Compatibility. Modular code, easy update, all browsers, any cloud deploy.	MVC architecture, AWS/Azure deployable, Chrome/Firefox/Edge/Safari, 90% test coverage
----------	-------	--	--